

JKBOSE 2026

CLASS 11TH

Photosynthesis in Higher Plants

LECTURE-06

BOTANY

ARSHID BHAT SIR

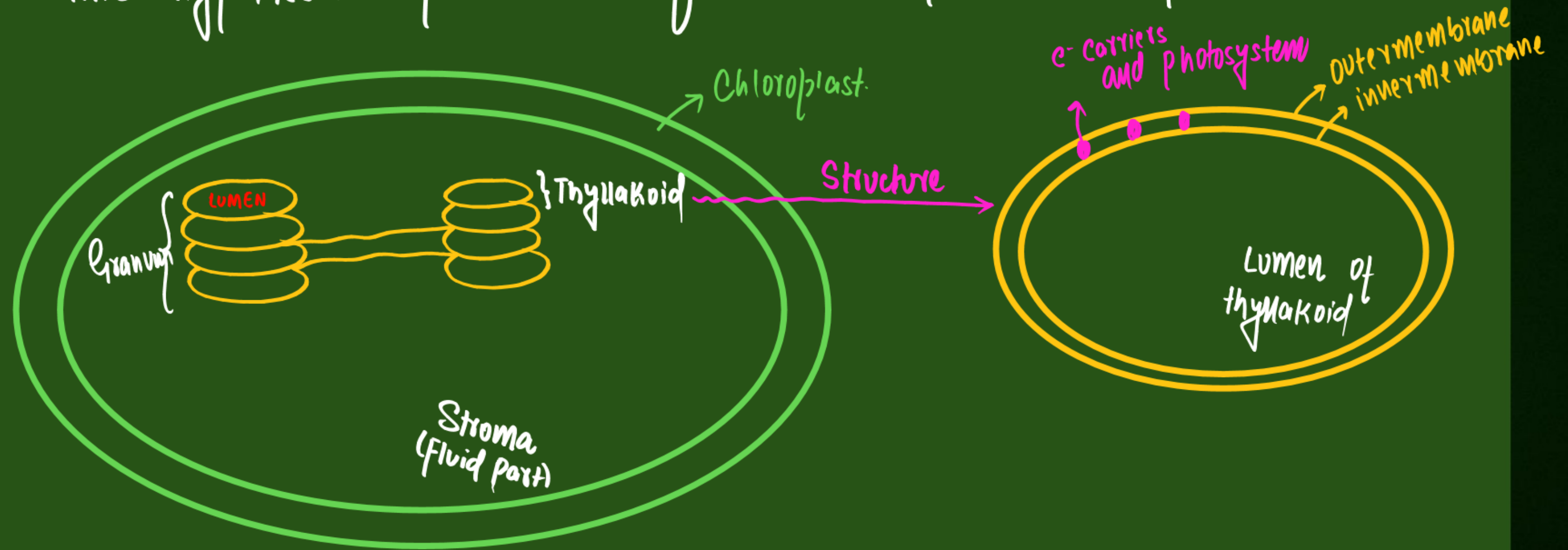


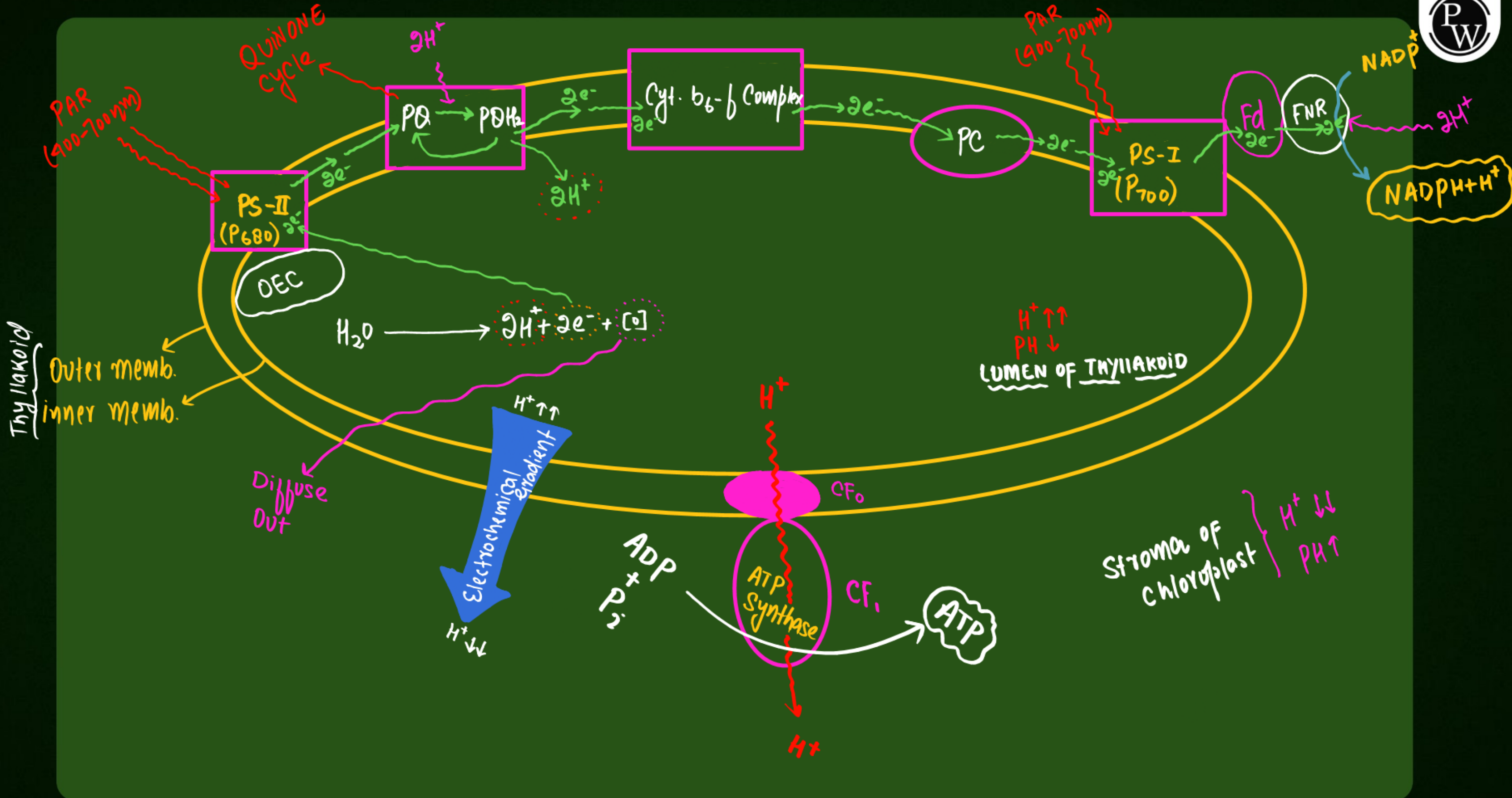
Topics To Covered

- 1** Chemiosmotic Hypothesis
- 2** Dark Reaction → introduction

Chemiosmotic Hypothesis

- Given by Peter Mitchell in 1961.
- This hypothesis explains the formation of ATP in chloroplast.





$\hat{=}$ C_{F_0} $\rightarrow$ It is a transmembrane protein which allows facilitated diffusion of H^+ from lumen of thylakoid to Stroma of Chloroplast.

C_{F_1} $\rightarrow$ It protrudes outside the thylakoid membrane towards Stroma of Chloroplast. It poses an enzyme — ATP Synthase.

* $\hat{=}$ Requirements for Chemiosmosis:

- a. Membrane (Thylakoid membrane).
- b. Proton gradient across the membrane.
- c. Proton pump (which facilitates diffusion of protons).
- d. ATP Synthase Enzyme.

Dark Reaction

- It is also called as Biosynthetic phase because here Glucose is formed.
- Site: Stroma of Chloroplast
- Requirements for Dark rxn:
 - a. ATP & $\text{NADPH} + \text{H}^+$ (from light rxn.)
 - b. CO_2
 - c. H_2O
- Dark rxn. is also called as Blackman's rxn. because it was first discovered by Blackmann.
- Dark rxn. is Enzymatically Controlled and thus temperature dependent.
- In Dark rxn., CO_2 is fixed and then reduced into Glucose.

- The technique used for studying various steps of Dark reaction was radioactive tracer technique by using ^{14}C autoradiography. For this, the organism used was Chlorella.

↓
• Unicellular green alga.
• Non-motile.

- Different photosynthetic organisms perform Dark rxn. differently.

- 03 different ways of Dark rxn.:

a. C_3 pathway → It involves C_3 -cycle / Calvin Cycle only.

b. C_4 pathway
c. CAM pathway } → involves both C_3 and C_4 cycle.

NOTE

- C_3 cycle / Calvin cycle is universal because it operates in all the photosynthetic plants (C_3 plant, C_4 plant, CAM plants).

QUESTIONS

Which one of the following products **diffuses out** of chloroplast during photosynthesis?
(2024 Re)

- A** ADP
- B** NADPH
- C** O_2
- D** ATP

QUESTIONS



NADP^+ is reduced to NADPH in:

(1988)

- A** PS I
- B** PS II
- C** Calvin cycle
- D** Non-cyclic photophosphorylation

HOMework

Revision:





Thank
You